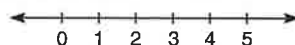


Lesson 6: Decimals and Fractions on the Number Line

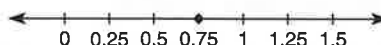
Key Ideas

- A number line is a picture that helps you compare the sizes of numbers. Numbers on the left are smaller than numbers on the right.
- Number lines can also be used to represent and compare decimals.
- Fractions and mixed fractions can also be represented on a number line.

On the GED® *Mathematical Reasoning Test* you may need to recognize or locate fractions, mixed fractions, or decimals on a **number line**. A number line represents numbers in order from least to greatest. As you move to the left along a number line, numbers decrease in value. As you move to the right, numbers increase in value.

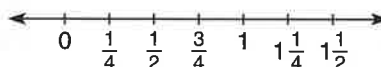


The arrows on the ends indicate that numbers continue forever in both directions. If you imagine zooming in on a portion of that number line, it might look like this:



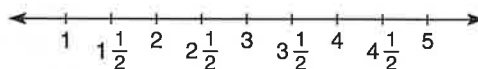
Thus, a number line can include decimals and whole numbers. For example, the point represents 0.75, which is greater than 0.5 and less than 1.

A number line can also represent fractions and mixed fractions in increasing order:



Use this question to practice working with number lines:

Example 1: John works five shifts per week. His boss asks him to spend exactly half of his time this week working on a specific project. How many work shifts will John devote to that project? Circle the answer on the number line below.



If you circled $2\frac{1}{2}$, you are correct. Multiply 5 by $\frac{1}{2}$ to determine that John will spend $2\frac{1}{2}$ shifts on the project.

ON THE GED® TEST

Questions on the Mathematical Reasoning Test may ask you to recognize a fraction or a decimal on a number line, or they may ask you to place a dot on a line to represent a fraction or decimal value.

Practice 6

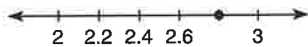
A. Choose the one best answer to each question.

1. What is the value of the point on the number line below?



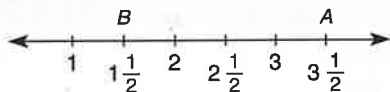
- A. 0
- B. $\frac{1}{2}$
- C. $\frac{2}{3}$
- D. $1\frac{1}{3}$

2. What is the value of the point on the number line below?



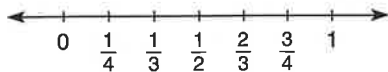
- A. 2
- B. 2.5
- C. 2.7
- D. 2.8

3. In the number line below, what is the value of A minus B?



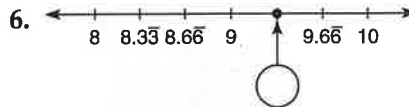
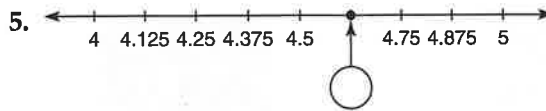
- A. 1
- B. $1\frac{1}{2}$
- C. 2
- D. $3\frac{1}{2}$

4. Angela baked 24 cookies and gave 16 of them to her neighbor. On the number line below, circle the fraction of Angela's cookies that she gave to her neighbor.



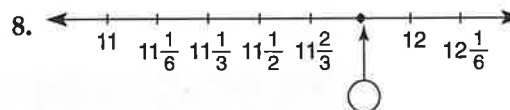
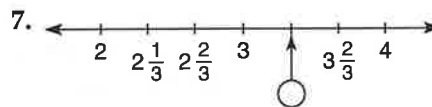
B. Write the decimal values of the points on the number lines.

For each of the number lines below, fill in the value of the point using decimals.



C. Write the fraction values of the points on the number lines.

For each of the number lines below, fill in the value of the point using mixed fractions.



Answers and explanations begin on page 663.

Lesson 7: Decimal and Fraction Calculator Skills

Several important calculator keys are used to work with decimals and fractions.

Key Ideas

- Fraction and decimal operations are entered in the same order as whole-number operations.
- On the TI-30XS MultiView™ calculator, use the decimal point key (.) for decimals and the fraction key, $\frac{n}{d}$, for fractions.
- To work with mixed fractions, use the 2nd function, $\frac{U}{d}$, which is in green over the fraction key.



Cursor keys move the cursor on the screen up, down, left, and right.

The **fraction key** is used to enter fractions; the **mixed number 2nd function** is in green above it.

The **toggle key** changes between equivalent fraction and decimal forms of a number.

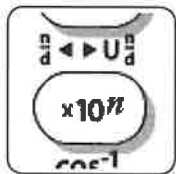
The **decimal point key** enters a decimal point.

Calculator **decimal operations** are performed in the same way that you use operations with whole numbers. You need to use the **decimal point key** $\cdot$ under the 2 in the white **numeric keypad**. Practice with these examples:

To solve this problem ...	Press these keys ...	The right-hand side of the display reads ...
$3.89 + 2.5$	$3.89 (+) 2.5 \text{ (enter)}$	6.39
$5.2 - 0.78$	$5.2 (-) .78 \text{ (enter)}$	4.42
0.9×15	$.9 (\times) 15 \text{ (enter)}$	13.5
$1.7 \div 2$	$1.7 (\div) 2 \text{ (enter)}$	0.85

GED® TEST TIP

A fraction answer may appear in the form of an improper fraction. In order to change to an equivalent mixed number, use the green 2nd function over this key:



You will use several calculator functions to work with fractions and **mixed fraction operations**. First practice entering fractions and converting to decimals.

Example 1: Reduce $\frac{56}{448}$ to lowest terms and then convert to a decimal.

- Clear the calculator.
- Press the $\frac{n}{d}$ button to enter a fraction. Enter 56 at the blinking cursor, in the numerator.
- Use the down cursor key ∇ to enter 448 in the denominator.
- Press enter to reduce the fraction to lowest terms, which appears on the right of your screen: $\frac{1}{8}$
- To express the fraction as a decimal, press the toggle button: $\leftrightarrow$. The decimal 0.125 appears on the right display.

On the Display

$$\frac{56}{448}$$

$$\frac{56}{448}$$

$$\frac{56}{448} \quad \frac{1}{8}$$

$$\frac{1}{8} \leftrightarrow 0.125$$

Now practice operations with **mixed fractions** using the *2nd function* key.

Example 2: A plastic pipe is to be cut into pieces measuring $1\frac{7}{8}$ feet. The original pipe was $20\frac{5}{8}$ feet long. How many pieces can be cut from the pipe?

	Keys to Press	On the Display
1. Clear the calculator		
2. Recognize that a mixed fraction is a green <i>2nd</i> function: over the key. Press the key and the key. Note both a whole number and a blinking fraction cursor on the display.		
3. Enter the number being divided first—the whole pipe: $20\frac{5}{8}$. Enter 20, then follow the direction of the onscreen arrow and press the right arrow button to move to the fraction cursors. Enter 5 and then use the down arrow to enter the 8. Exit the fraction by pressing the right arrow again.	20 5 8	
4. Press the division key. Then follow the same process with the second number: $1\frac{7}{8}$, starting with the <i>2nd</i> function:	 1 7 8	
5. Press the button for the solution. The answer 11 for 11 pieces appears on the right side of the screen.		

Practice 7

A. Solve the following problems using your calculator.

- $3.5 + 1.87 + 2.009$
- $3\frac{2}{3} - 1\frac{5}{12}$
- $\$25.35 \times 15$
- $10\frac{1}{2} \div \frac{1}{4}$
- Linda earns \$95 per day. If she works $\frac{8}{9}$ of a day, how much will she earn?
- Aaron bought a refrigerator for \$956.88. The price includes tax and interest charges. If he makes 12 equal monthly payments, how much will he pay each month?
- An insurance agent estimates the annual cost of insurance on a home by multiplying the sale price of the home by 0.0125. What will be the yearly cost of insurance on a home priced at \$118,000?
- In a recipe, the total liquid added to a mixture is $1\frac{1}{2}$ cups of water and $2\frac{3}{4}$ cups of chicken broth. How many cups of liquid are used in the recipe?
- A quilt costs \$84.99 and weighs 5.56 pounds. The shipping charge is \$1.20 per pound. To the nearest cent, what would be the shipping charge on the quilt?
 - \$5.56
 - \$6.67
 - \$8.26
 - \$10.20
- A state park contains 64 acres. A wild-life preserve makes up $\frac{3}{8}$ of the park. How many acres are in the preserve?
 - 8
 - 21
 - 24
 - 27

Answers and explanations begin on page 664.

Decimals and Fractions Practice Questions

Directions: You MAY use your calculator.

1. A wooden flooring strip is $20\frac{1}{2}$ inches long. If you cut off $4\frac{3}{4}$ inches from one end, what will be the new length of the strip in inches?
A. $16\frac{3}{4}$
B. $16\frac{1}{4}$
C. $15\frac{3}{4}$
D. $15\frac{1}{4}$
2. A box of cereal costs \$4.69. The package label says that the box contains 19 servings. What is the cost of 1 serving to the nearest cent?
A. \$0.02
B. \$0.25
C. \$0.47
D. \$2.46
3. How much would a computer system cost if it is priced as shown below?

Pay \$200 down and make 12 monthly payments of only \$98.85.

A. \$3586.20
B. \$2400.00
C. \$1386.20
D. \$1186.20
4. Unleaded gasoline sells for \$2.869 per gallon. How much would $10\frac{1}{2}$ gallons cost? Round your answer to the nearest cent.
A. \$13.40
B. \$28.69
C. \$30.12
D. \$301.20
5. Three packages weigh $1\frac{1}{2}$ pounds, $4\frac{3}{4}$ pounds, and $2\frac{3}{10}$ pounds. What is the average weight, in pounds, of the packages? (*Hint:* Add the weights, then divide by the number of packages.)
A. 2.14
B. 2.85
C. 4.75
D. 8.55
6. Gina is paid \$8 an hour. If she earned \$258 in 1 week, how many hours did she work?
A. $34\frac{1}{2}$
B. $32\frac{1}{3}$
C. $32\frac{1}{4}$
D. $32\frac{1}{5}$
7. A developer plans to build homes on $20\frac{1}{2}$ acres. She estimates that $6\frac{1}{4}$ acres will be used for roads. The remaining land will be divided into $\frac{1}{4}$ -acre lots. How many lots can the subdivision include?
A. 7
B. 57
C. 81
D. 107
8. A school buys 1000 white-board markers. Below is the price per marker for two brands. How much did the school save by buying Brand A instead of Brand B?
Brand A: \$0.27 each
Brand B: \$0.36 each
A. \$0.09
B. \$0.90
C. \$9.00
D. \$90.00

Questions 9 and 10 refer to the following information.

Madison Small Animal Clinic Scheduling Guidelines	
New-Patient Appointment	$\frac{3}{4}$ hr
Immunizations	$\frac{1}{4}$ hr
Routine Physical	$\frac{1}{3}$ hr
Dental Scaling	$\frac{3}{4}$ hr
Sick Animal Visit	$\frac{1}{2}$ hr
Serious-Injury Visit (includes X-rays)	$1\frac{1}{4}$ hr

9. Ray is a veterinarian at the small-animal clinic. He has four appointments scheduled for Monday morning: two new-patient appointments, a serious-injury visit, and a dental scaling. In hours, how much time should these appointments take?

- A. $2\frac{3}{4}$
- B. $3\frac{1}{2}$
- C. $3\frac{3}{4}$
- D. 4

10. Jennifer works $3\frac{1}{2}$ hours each morning at the clinic. How many routine physicals could she complete in one morning?

- A. 4
- B. 9
- C. 10
- D. 11

11. A minor-league baseball stadium has 6000 seats. On Beach Towel Night, the stadium sold 5500 of its available seats. What fraction of the seats were sold?

- A. $\frac{5}{6}$
- B. $\frac{8}{9}$
- C. $\frac{9}{10}$
- D. $\frac{11}{12}$

12. The City Center parking garage charges \$3.50 for the first hour and \$1.25 for each additional $\frac{1}{2}$ hour. How much would it cost to park at the garage for $2\frac{1}{2}$ hours?

- A. \$4.75
- B. \$6.25
- C. \$7.25
- D. \$9.75

13. Susan scheduled 84 appointments for patients at a hospital outreach clinic. Only 56 patients kept their appointments. What fraction of the scheduled appointments were kept?

Write your answer on the line below.

14. This portion of a gas bill compares a household's natural gas usage for December of this year and last year.

Gas Bill Comparison Average Daily Usage			
This Year		Last Year	
Dec	3.13 therms	Dec	3.97 therms

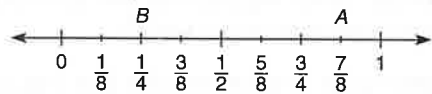
How many more therms of natural gas did the household use in December of last year than in December of this year? Express your answer as a decimal.

Write your answer on the line below.

15. Jim and Carl have until 1 p.m. to load 250 boxes. By 12:30 p.m., 175 of the boxes are loaded. What fraction of the boxes has NOT been loaded?

A. $\frac{1}{5}$
 B. $\frac{3}{10}$
 C. $\frac{3}{5}$
 D. $\frac{7}{10}$

16. On the number line below, what is the value of A minus B ?



A. $\frac{1}{4}$
 B. $\frac{5}{8}$
 C. $\frac{7}{8}$
 D. $1\frac{1}{8}$

17. Scott is driving about 380 miles from Los Angeles to San Francisco. He plans to cover $\frac{3}{4}$ of the distance before noon. How many miles does he plan to drive before noon?

A. 285
 B. 254
 C. 126
 D. 95

18. A cookie recipe calls for $1\frac{2}{3}$ cups of sugar. If you wanted to make half the quantity shown in the recipe, how many cups of sugar would you use?

A. $\frac{2}{5}$
 B. $\frac{2}{3}$
 C. $\frac{5}{6}$
 D. $1\frac{1}{6}$

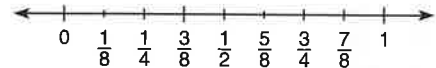
Question 19 is based on the following information.

Carbide Steel Drill Bits		
Description	Size (inches)	Price
Cutter	$\frac{9}{16}$	\$6.19
Core Box	$\frac{5}{32}$	\$16.40
Classic	$\frac{3}{8}$	\$17.85
Bevel	$\frac{1}{2}$	\$10.50

19. Which of the following shows the drill bits arranged in order from least to greatest in size?

A. cutter, bevel, classic, core box
 B. core box, bevel, classic, cutter
 C. bevel, classic, cutter, core box
 D. core box, classic, bevel, cutter

20. Joe is going to order a pizza. He will eat at least $\frac{1}{2}$ of it. If he's very hungry, he might eat as much as $\frac{7}{8}$ of it. Place two points on the number line below to represent the minimum fraction of the pizza Joe might have left over and the maximum fraction of the pizza Joe might have left over.



21. A project should take no more than 60 hours. If John can spare 7.5 hours per day to work on the project, what is the maximum number of days it will take him to finish?

A. 6
 B. 7
 C. 8
 D. 9

22. How many miles would a map distance of $\frac{5}{8}$ inch represent if 1 in. = 240 mi?

A. 40
 B. 130
 C. 150
 D. 200

23. A survey shows that $\frac{2}{3}$ of all homeowners have a pet. Of those, $\frac{3}{4}$ have either a dog or cat. Of the homeowners surveyed, what fraction has either a dog or a cat?

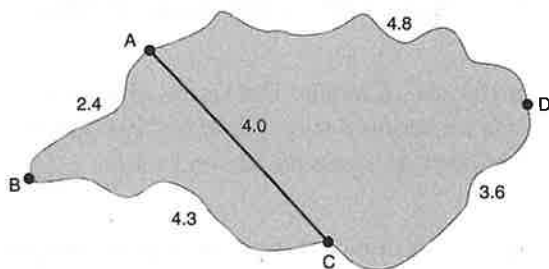
A. $\frac{1}{2}$
B. $\frac{3}{4}$
C. $\frac{5}{7}$
D. $\frac{6}{7}$

24. At Wyman Shipping, 140 employees work during the day shift. At night, the crew is $\frac{2}{5}$ the size of the day shift. How many workers are scheduled to work the night shift?

A. 28
B. 56
C. 70
D. 94

Question 25 refers to the following map.

The map below shows the distance, in miles, between four stores.



25. Maya drives a van that delivers supplies to each of the stores. On Friday, she traveled the following route.

Store A to Store B
Store B to Store C
Store C to Store D
Store D to Store C
Store C to Store A

How many miles did she drive in all?

A. 13.9
B. 15.1
C. 17.9
D. 19.1

26. From a wooden dowel $12\frac{1}{2}$ feet long, Jamie cut two pieces, each $3\frac{3}{4}$ feet long. How long, in feet, is the remaining piece?

A. $8\frac{3}{4}$
B. $7\frac{1}{2}$
C. $6\frac{1}{2}$
D. 5

27. Luis bought 20 shares of stock, priced at \$26.38 per share. He also paid an \$8 transaction fee. How much did he pay for the shares and the fee?

A. \$687.60
B. \$535.60
C. \$527.60
D. \$519.60

28. A mat board is 60 inches wide. How many strips measuring 0.75 inches wide can be cut from the board? (Assume no waste from the cuts.)

Write your answer on the line below.

29. Of his take-home pay each month, Jerry spends $\frac{1}{6}$ on a car payment and $\frac{1}{4}$ on food. What fraction of his take-home pay is left after paying for these two items?

Write your answer on the line below.

Answers and explanations begin on page 664.

Ratio, Proportion, and Percent

Lesson 1: Ratio and Proportion

Key Ideas

- A ratio has two terms and can be written in words, with a colon, or as a fraction.
- Ratios are simplified by reducing to lowest terms.
- The terms in a ratio can have different labels.

GED® TEST TIP

The terms in a ratio must be written in the order given by the words of the question. Don't be fooled by an answer choice on a multiple-choice question that reverses the order of the numbers.

Ratio

A **ratio** compares two numbers. You can write a ratio using the word *to*, using a colon (:), or using fraction form.

Example 1: A softball pitcher strikes out four batters for every one batter that she walks. What is the ratio of strikeouts to walks?

Always write the numbers in the ratio in the same order in which they appear in the question.

4 to 1	4:1	$\frac{4}{1}$
--------	-----	---------------

Ratios are similar to fractions. They have two terms, and they can be simplified by reducing to lowest terms.

Example 2: Frank manages a small drugstore. During a two-hour period, he counts 25 cash sales and 15 credit card sales. What is the ratio of credit card to cash sales?

1. Write the ratio as a fraction with the terms in the correct order: credit card to cash sales.

$\frac{\text{credit card sales}}{\text{cash sales}} = \frac{15}{25}$
--
2. Reduce to lowest terms. The ratio of credit card to cash sales is **3 to 5**.

$\frac{15 \div 5}{25 \div 5} = \frac{3}{5}$

There are some fraction rules that ratios do not follow. Do not change a ratio that is an improper fraction to a mixed number. Also, if a ratio in fraction form has a denominator of 1, do not write it as a whole number. Leave it in fraction form.

Another important difference is in the use of labels. The terms in a fraction have the same unit labels: $\frac{5}{6}$ of a pie means 5 slices out of 6 slices. Ratios may have different labels: The sale advertised 6 cans for \$1, a 6:1 ratio.

To write a ratio, you may need to perform one or more basic operations to find one of the terms.

Example 3: A football team won 12 games and lost 8. There were no tied games. What is the ratio of games won to games played?

1. The problem does not tell you the number of games played. Add the games won to the games lost to find the games played.

$12 \text{ won} + 8 \text{ lost} = 20 \text{ played}$

2. Write the ratio in the correct order and simplify. **The team won 3 games for every 5 games it played, a 3:5 ratio.**

$\frac{\text{games won}}{\text{games played}} = \frac{12}{20} = \frac{12 \div 4}{20 \div 4} = \frac{3}{5}$
--

Practice 1.1

A. Write each ratio as a fraction in lowest terms.

1. Stan made 24 sales in 6 hours. What is the ratio of sales to hours?
2. Carol's monthly take-home pay is \$1500. She spends \$250 a month on food. What is the ratio of food costs to take-home dollars?
3. A toy rocket travels 180 ft in 15 sec. What is the ratio of feet to seconds?
4. At Phil's work, there are 12 part-time workers and 18 full-time workers. What is the ratio of part-time workers to total workers?
5. Juanita drove 336 miles on 14 gallons of gasoline. What is the ratio of miles to gallons?
6. Lynn estimates that a roofing job will cost \$1500. Bo estimates that the same job will cost \$2400. What is the ratio of Lynn's estimate to Bo's estimate?
7. A basketball player attempted 32 free throws and made 20. What is the ratio of free throws made to free throws missed?
8. There are 10 men and 14 women in Kathleen's math class. What is the ratio of women to the total number of students in the class?
9. To paint his apartment, Alex bought 6 gallons of paint to cover 1440 square feet. What is the ratio of square feet to gallons of paint?

B. Choose the one best answer to each question.

Questions 10 through 12 refer to the following information.

Three candidates are running for mayor. Below are the results of a survey of 600 registered voters.

Candidate	Number of Supporters
Stothard	220
Mesa	180
Newmark	50
Undecided	150

10. What is the ratio of Mesa's supporters to Stothard's supporters?
A. 9:11
B. 11:9
C. 11:20
D. 20:11
11. What is the ratio of voters who prefer Mesa to the total number surveyed?
A. 3 to 7
B. 3 to 10
C. 3 to 13
D. 11 to 30
12. What is the ratio of undecided voters to voters who have made a decision?
A. $\frac{1}{4}$
B. $\frac{1}{3}$
C. $\frac{3}{1}$
D. $\frac{4}{1}$
13. Soan made a \$400 down payment on a washer and dryer that cost a total of \$1200. What is the ratio of the amount Soan has paid to the amount he still owes?
A. 1 to 4
B. 1 to 3
C. 1 to 2
D. 2 to 3
14. A team played 77 games and won 56 of them. There were no tied games. What is the ratio of wins to losses?
A. 3:8
B. 8:11
C. 11:8
D. 8:3

Answers and explanations begin on page 664.

Proportion

A **proportion** is an equation that shows that two ratios are equal. The **cross products** in a true proportion are equal. In other words, when you multiply diagonally across the equals sign, the products are equal.

Example 4: The directions on a can of powdered drink mix say to add 3 cups of water to every 2 scoops of drink mix. Matt adds 12 cups of water to 8 scoops of drink mix. Did he make the drink correctly?

1. Write a proportion, making sure the terms of the ratios are in the same order.
$$\frac{\text{cups}}{\text{scoops}} = \frac{3}{2} = \frac{12}{8}$$
2. Cross multiply and compare the products. Since the products are the same, the ratios are equal. **Matt made the drink correctly.**
$$3 \times 8 = 24$$
$$2 \times 12 = 24$$

In most proportion problems, you are asked to solve for a missing term.

Example 5: A map scale says that 2 inches = 150 miles. What actual distance would a map distance of 5 inches represent?

1. Write a proportion with both ratios in the same form: inches to miles. The variable x represents the unknown distance.
$$\frac{\text{inches}}{\text{miles}} = \frac{2}{150} = \frac{5}{x}$$
2. Locate the term in the first ratio that is diagonal from the known term in the second ratio. Cross multiply.
$$\frac{2}{150} = \frac{5}{x}$$
$$150 \times 5 = 750$$
3. Divide the result by the remaining known term to find the value of x .
$$750 \div 2 = 375 \text{ miles}$$

Some proportion problems ask you to find a **rate**. A rate compares a quantity to 1. When a rate is written in fraction form, its denominator is always 1. In word form, rates are often expressed using the word *per*.

Example 6: Connie drove 276 miles on 12 gallons of gasoline. How many miles per gallon did she get on the trip?

1. Gas mileage is one kind of rate. You need to find how many miles Connie drove on one gallon of gasoline.
$$\frac{\text{miles}}{\text{gallons}} = \frac{276}{12} = \frac{x}{1}$$
2. Solve.
$$276 \times 1 = 276$$
$$276 \div 12 = 23 \text{ miles per gallon}$$

Using your calculator, you can solve proportion problems in one series of calculations.

Example 7: Find the value of x in the proportion $\frac{6}{16} = \frac{21}{x}$.

You need to multiply 16 and 21, then divide by 6. On the TI-30XS MultiView™ calculator:

1. Press $16 \times 21 \div 6$.
2. Then press enter . The right side of the display will read 56.

The missing term is 56.

NOTE: When working a problem, ask yourself if it can be solved using proportion. This may be possible when two quantities are compared or when three values are given and you are asked to find a fourth.

Practice 1.2

A. Solve for the missing term in each proportion problem. You MAY use your calculator for questions 9 through 16. NOTE: Answers will not always be whole numbers.

1. $\frac{2}{3} = \frac{x}{18}$

5. $\frac{4}{\$212} = \frac{7}{x}$

9. $\frac{20}{2.5} = \frac{100}{x}$

13. $\frac{3}{19} = \frac{x}{114}$

2. $\frac{3}{5} = \frac{27}{x}$

6. $\frac{25}{6} = \frac{400}{x}$

10. $\frac{\$5.96}{2} = \frac{x}{3}$

14. $\frac{9}{\$80.10} = \frac{x}{\$284.80}$

3. $\frac{6}{5} = \frac{3}{x}$

7. $\frac{7}{30} = \frac{x}{9}$

11. $\frac{12}{5} = \frac{3}{x}$

15. $\frac{\$26.00}{4} = \frac{x}{7}$

4. $\frac{15}{2} = \frac{x}{8}$

8. $\frac{0.5}{12} = \frac{3}{x}$

12. $\frac{4}{60} = \frac{2.5}{x}$

16. $\frac{24}{96} = \frac{7}{x}$

B. Choose the one best answer to each question. You MAY use your calculator for questions 20 through 22.

17. A store is advertising the following sale:

Tomato Soup 4 cans for \$0.98

To the nearest cent, how much would five cans of tomato soup cost?

- A. \$0.25
- B. \$1.23
- C. \$2.45
- D. \$4.90

18. The Bay City Cardinals have won 5 out of 8 games. At the same rate, how many games will they have to play to win 60 games?

- A. 180
- B. 120
- C. 96
- D. 12

19. Carla drove her truck 414 miles on 18 gallons of gasoline. How many miles did she drive per gallon?

- A. 18
- B. 23
- C. 74
- D. 95

20. The scale on a map reads, "2 cm = 150 km." How many kilometers would be represented by a distance of 4.6 centimeters?

- A. 300
- B. 345
- C. 690
- D. 1380

21. Two ingredients in a recipe are $2\frac{1}{2}$ cups of flour and $1\frac{1}{2}$ cups of sugar. If June keeps the proportion the same, how many cups of flour should she add to 4 cups of sugar?

- A. $6\frac{2}{3}$
- B. 6
- C. 5
- D. $3\frac{3}{4}$

22. Claudia drove 155 miles in 2.5 hours. Which of the following expressions could be used to find how many miles she can drive in 7 hours?

- A. $155 \times 7 \div 2.5$
- B. $2.5 \times 7 \div 155$
- C. $155 \times 2.5 \div 47$
- D. $7 \times 2.5 \times 155$

Answers and explanations begin on page 665.

Lesson 2: Percents

Key Ideas

- The percent symbol (%) means “out of 100.”
- The three elements of a percent problem are the base, the part, and the rate.
- Percent problems can be solved by writing a proportion that has a denominator of 100.

Percent means “per hundred” or “out of one hundred.” For example, if you have \$100 and you spend \$25, you spent \$25 out of \$100, or 25% of your money.

Since percent is a way of showing part of a whole, it has much in common with fractions and decimals. To convert a percent to a fraction, write the percent over 100 and reduce. To convert percents to decimals, drop the percent symbol and move the decimal point two places to the left.

Percent to Fraction

$$25\% = \frac{25}{100} = \frac{1}{4}$$

Percent to Decimal

$$25\% = .25 = 0.25$$

In any percent problem, there are three elements: the base, the part, and the rate. The **base** is the whole quantity, or amount, that the problem is about. The **part** (also called a **percentage**) is a portion of the base. The **rate** is a number followed by the percent symbol (%).

Example 1: At a restaurant, Janice’s bill is \$20. She gives the waiter a tip of \$3, which is 15% of her bill. Identify the base, part, and rate in this situation.

The entire bill of \$20 is the base. The \$3 tip is part of the base, and the rate is 15%.

One way to think of a percent problem is as a proportion. In Example 1, there are two ratios. The \$3 tip is part of the \$20 total bill, and 15% is the same as $\frac{15}{100}$. Since the two ratios are equal, they can be written as a proportion.

$$\frac{\text{part}}{\text{base}} = \frac{3}{20} = \frac{15}{100}$$

Cross multiply to prove the ratios are equal.

$$\begin{aligned} 20 \times 15 &= 300 \\ 3 \times 100 &= 300 \end{aligned}$$

You can solve percent problems by setting up a proportion and solving for the missing elements. Just remember to express the percent as a number over 100.

Example 2: At a plant that manufactures lighting fixtures, it is expected that about 2% of the fixtures assembled each day will have some type of defect. If 900 fixtures are completed in one day, how many are expected to be defective?

1. Write a proportion. Remember that 2% means 2 out of 100: Use the variable x to stand for the number of defective fixtures.

$$\frac{\text{part}}{\text{base}} = \frac{\text{rate}}{100}$$

$$\frac{x}{900} = \frac{2}{100}$$

2. Solve for x . Cross multiply and divide by the remaining number. **The company can expect about 18 defective fixtures.**

$$\begin{aligned} 900 \times 2 &= 1800 \\ 1800 \div 100 &= 18 \end{aligned}$$

GED® TEST TIP

The word “of” often comes before the base in a percent problem. For example, if you are asked to find 75% of 250, you know that 250 is the base.

Practice 2

A. For each situation, identify and label the base, part, and rate.

- Victor owes his uncle \$1000. Recently, he gave his uncle \$200. The payment was 20 percent of the money he owes.
- On a test with 80 problems, Sophie got 72 problems right. In other words, she answered 90 percent of the problems correctly.
- The Kang family made a down payment of \$2,740 on a new car. The down payment was 20 percent of the purchase price of \$13,700.
- Zoe's take-home pay each month is \$2000. She spends \$500 on rent each month, which is 25 percent of her take-home pay.
- This year, Rafael has 60 regular customers, which is 150 percent of the 40 regular customers he had last year.
- Kayla bought a dress for \$38. She paid \$3.23 in sales tax. The sales tax rate in her state is 8.5 percent.
- Misako's employer withholds 15 percent of her salary each paycheck for taxes. Misako earns \$900 each week, and her tax withholding is \$135.
- Harrison got a 10 percent raise. Before the raise, his hourly wage was \$10.70. Now he earns an additional \$1.07 per hour.
- Kim Industries has 800 employees. Of those, 200 workers, or 25 percent, work part-time.
- In an election, 5,000 of the 12,500 registered voters actually voted. Only 40 percent of the registered voters actually voted.

B. Choose the one best answer to each question. Use the proportion $\frac{\text{part}}{\text{base}} = \frac{\text{rate}}{100}$ to solve each problem.

Questions 11 and 12 refer to the following information.

A local newspaper printed the following high school basketball standings:

Team	Wins	Losses
Fairfax	9	3
Hamilton	8	4
Bravo	6	6
Mountain View	4	8
Lincoln	3	9

11. Which of the following expressions could be used to find what percent of its total games Fairfax has won?

- $\frac{9 \times 100}{12}$
- $\frac{3 \times 100}{12}$
- $\frac{12 \times 100}{9}$
- $\frac{6 \times 100}{12}$

12. What percent of its games did Bravo win?

- 100%
- 75%
- 60%
- 50%

13. A jacket with a price tag of \$128 is on a rack with the following sign:

All Items:
25% off marked price
Discount taken at register

By how much will the price be reduced when the jacket is taken to the register?

- \$4
- \$25
- \$32
- \$96

Answers and explanations begin on page 665.

Lesson 3: Using the Percent Formula

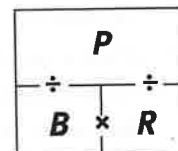
Key Ideas

- You can use a formula to solve percent problems: $\text{Base} \times \text{Rate} = \text{Part}$.
- To change a percent to a decimal, drop the % sign and move the decimal point two places to the left.
- To change a decimal to a percent, move the decimal point two places to the right and add the % sign.

Solving for Part

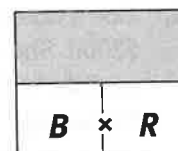
You have seen how to use proportion to solve percent problems. You can also solve percent problems using the formula $\text{Base} \times \text{Rate} = \text{Part}$.

Study the diagram at the right to learn how to use the formula. To use the diagram, cover the element you need to solve for: $B = \text{Base}$, $R = \text{Rate}$ (percent), and $P = \text{Part}$. Then perform the operation that connects the remaining elements.



Example 1: A company offers its employees two health plans. In a recent newsletter, the personnel department stated that 70% of the employees chose Plan A. If the company has 320 workers, how many chose Plan A?

- The rate is 70%, and the base is 320, the total number of workers. You need to solve for the part. Using the diagram, cover P for part. You can see that you need to multiply to solve the problem.



- Change the percent to a decimal and multiply.
Out of 320 workers, 224 chose Plan A.

$$70\% = 0.7$$

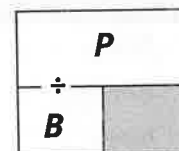
$$320 \times 0.7 = 224$$

Solving for Rate

Rewrite the percent formula to solve for rate. Use the formula $\text{Part} \div \text{Base} = \text{Rate}$. You can use the diagram to help you remember the formula.

Example 2: A computer system is regularly priced at \$1600. On Friday, the manager reduced the price by \$640. By what percent did the manager discount the computer system?

- The base is \$1600, the regular price. The part is \$640, the amount the price was reduced. You are asked to find the rate of the discount. Cover R for rate (percent). You need to divide the part by the base to solve the problem.



- Divide 640 by 1600.

- Convert the decimal answer to a percent by moving the decimal point two places to the right and adding the percent sign. The price reduction was a 40% discount.

$$\begin{array}{r} 0.4 \\ 1600 \overline{)640.0} \\ \underline{640.0} \\ 0.4 = 40\% = 40\% \end{array}$$

Always ask yourself whether your answer seems reasonable. For example, you know that 40% is a little less than $\frac{1}{2}$, and $\frac{1}{2}$ of \$1600 is \$800. Since \$640 is a little less than \$800, it is a reasonable answer.

GED® TEST TIP

When you take the Mathematical Reasoning Test, make a quick sketch of the percent diagram on the wipe-off board to help you analyze percent problems.

Practice 3.1

A. Solve. You MAY use a calculator for questions 9 through 16.

1. What is 20% of \$25?
2. Find 90% of 200.
3. What is 35% of 400?
4. What percent is 19 out of 20?
5. 42 is what percent of 168?
6. What percent is \$18 out of \$600?
7. Find $33\frac{1}{3}\%$ of 51. (*Hint: $33\frac{1}{3}\% = \frac{1}{3}$*)
8. What is 125% of \$48?
9. 240 is what percent of 120?
10. What percent is 3 out of 60?
11. \$52 is what percent of \$650?
12. Find $8\frac{1}{2}\%$ of \$46.
13. \$0.65 is what percent of \$10.00?
14. Find 28% of \$1300.
15. What percent is 2.5 out of 4?
16. Find $66\frac{2}{3}\%$ of 108. (*Hint: $66\frac{2}{3}\% = \frac{2}{3}$*)

B. Choose the one best answer to each question. You MAY use your calculator.

17. Pat called 120 customers to offer a software upgrade. Of those he called, 72 purchased the upgrade. What percent agreed to the purchase?
A. 40%
B. 48%
C. 60%
D. $66\frac{2}{3}\%$
18. Douglas received a 6 percent raise. If his old monthly salary was \$2,250, what is his monthly salary now? (*Hint: Find the amount of the raise. Then add the raise to the previous monthly salary.*)
A. \$2,256
B. \$2,385
C. \$3,600
D. \$13,500
19. At a restaurant, Levy's total bill is \$46. If he wants to tip 15 percent, how much should he leave as a tip?
A. \$690.00
B. \$31.00
C. \$15.00
D. \$6.90
20. The following advertisement for sporting goods appeared in the newspaper. What percent of the original price is the sale price?

Little League Package
Magnum bat, tote bag, and youth cleats
Only \$45.50
Originally \$65

A. 20%
B. 31%
C. 44%
D. 70%
21. Lydia pays \$3 sales tax on a \$50 purchase. Which of the following expressions could be used to find the sales tax rate in her state?
A. $\frac{\$3 \times 100}{\$50}$
B. $\frac{\$3 \times \$50}{100}$
C. $\$3 \times \50×100
D. $\$3 \div \50

Answers and explanations begin on page 665.

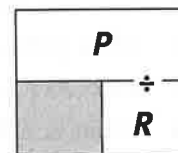
Solving for Base

Some problems on the *Mathematical Reasoning Test* may require you to solve for the base in a percent situation. Remember, the base represents the whole item or group. Read each situation carefully to figure out which element is missing. Then choose the correct method for solving the problem.

Example 3: In a math class, 75% of the students got at least a B grade on the final exam. If 18 students got at least a B, how many students are in the class?

1. Analyze the situation. The 18 students are part of the larger class. You know that the 18 students are 75% of the whole group, so 75% is the rate and the base is unknown.

Use the diagram. Cover *B* for base. You need to divide the part by the rate to solve the problem.



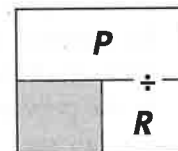
$$\begin{array}{r} 24 \\ 0.75 \overline{)18.00} \\ \underline{150} \\ 300 \\ \underline{300} \end{array}$$

2. Convert the rate to a decimal (75% = 0.75) and divide. There are **24 students** in the class.

Most of the time, we work with percents that are less than 100%. When a percent is less than 100%, the part is less than the base. However, it is possible to have a situation in which the part is greater than the base. When this occurs, the percent will be greater than 100%.

Example 4: The workforce at Eastland Inc. is growing rapidly. The number of employees this year is 225% of the number last year. If there are 135 employees this year, how many employees did the company have last year?

1. The base is the number of employees the company had last year. This year's number is a percent of last year's number. Therefore, the rate is 225%, the part is 135, and the base is unknown.



2. Convert 225% to a decimal. Drop the % sign and move the decimal point two places to the left.

$$225\% = 2.25 = 2.25$$

3. Divide the part (135) by the rate (2.25). Last year, there were only **60 employees**.

$$\begin{array}{r} 60 \\ 2.25 \overline{)135.00} \\ \underline{135.0} \end{array}$$

There is often more than one way to approach the solution to a problem. Both the percent formula presented in this lesson and the proportion method from Lesson 1 can be used to solve Example 4.

Formula method: $\frac{135}{2.25}$ Proportion method: $\frac{135 \times 100}{225}$

If you evaluate both methods using a calculator, both expressions equal 60, the correct solution.

NOTE: Don't begin calculations before you completely analyze a situation. Every percent problem has three elements. Make sure you know which one is missing before you multiply or divide.

Practice 3.2

A. Find the missing element in each set. You MAY NOT use a calculator.

1. \$35 is 20% of what amount?
2. 5% of what number is 14?
3. 3.2 is 50% of what number?
4. \$170 is 85% of what amount?
5. 24 is 80% of what number?
6. \$105 is 125% of what amount?
7. 190 is 95% of what number?
8. What number is 15% of 60?
9. 90% of \$15 is what number?
10. \$42 is what percent of \$168?
11. \$150 is 200% of what amount?
12. 15% of \$62 is what amount?
13. 9 is 1% of what number?
14. What percent is 126 of 140?
15. 65% of \$1200 is what amount?
16. 5% of an amount is \$156. What is the amount?
17. $2\frac{1}{2}\%$ of a number is 100. What is the number?
18. What percent is \$15.60 of \$156.00?

B. Choose the one best answer to each question. You MAY use your calculator.

19. Kevin's total payroll deductions are 30 percent of his earnings. If his deductions add up to \$369 for a two-week period, how much were his earnings for the period?

A. \$110.70
B. \$123.00
C. \$1,230.00
D. \$11,070.00

20. A city council established the following budget to improve public transportation.

	Project Budget
Salaries	50%
Office lease	35%
Equipment	6%
Supplies	2%
Miscellaneous	7%

If \$72,000 is allotted for equipment, what is the total budget for the project?

A. \$432,000
B. \$940,000
C. \$1,200,000
D. \$120,000,000

21. Jack earns a 5 percent commission on each sale. If he is paid a \$160 commission, which of the following expressions could be used to find the amount of the sale?

A. $\frac{5 \times 100}{160}$
B. $\frac{160 \times 100}{5}$
C. $\frac{5 \times 160}{100}$
D. $5 \times 100 \times 1.60$

22. American Loan Company mailed 3600 customers an application for a new credit card. Only 20 percent of the customers returned the application. How many customers returned the application?

A. 72
B. 180
C. 720
D. 180,000

Answers and explanations begin on page 665.

Lesson 4: Percent Calculator Skills

Key Ideas

- One way to solve percent problems on the calculator is to change the percent to a decimal or a fraction.
- On the TI-30XS MultiView™ calculator, you can use the percent 2nd function to find the part and the base.
- To create a percent, use the 2nd function, %, which is in green over the left parenthesis key.

By changing a percent to either a fraction or decimal, you can use a calculator to solve percent problems.

Example 1: What is 25% of 120? Try both decimals and fractions.

Change 25% to the decimal (.25), multiply times 120, and press **(enter)**. 30 is on the right of the display.

On the Display

.25*120	30
---------	----

Change 25% to the fraction $\frac{25}{100}$ using the **($\frac{\Box}{\Box}$)** key, multiply times 120, and press **(enter)**. 30 is on the right of the display.

$\frac{25}{100}$ *120	30
-----------------------	----

Using either decimals or fractions, you can find that 30 is 25% of 120.

When you use the percent 2nd function, you don't have to convert the percent to a fraction or decimal. Practice this function below to multiply to find the **part** when you are given the rate (percent).

Example 2: What is 65% of \$360?

Keys to Press

On the Display

1. Type the base, 360, and press the multiplication sign.

360 **($\times$)**

360*	
------	--

2. Type the rate (65). Press the 2nd function key and engage the percent 2nd function—over the **($\frac{\Box}{\Box}$)** key. Then press **(enter)**.

65
(2nd) **($\frac{\Box}{\Box}$)** **(enter)**

360*65%	234
---------	-----

The amount **\$234** is 65% of \$360.

You can use the percent function to divide to find the **base** when given the rate.

GED® TEST TIP

Practice using the delete key when you make a mistake in a problem. That will erase your latest entry. The clear key will erase the entire entry.

Example 3: Ned paid \$150 for a stereo. The amount Ned paid was 20% of the original cost. What was the original cost of the stereo?

Keys to Press

On the Display

1. Enter the base, 150, and press the division sign.

150 **($\div$)**

150÷	
------	--

2. Enter 20 and press the 2nd function key and the **($\frac{\Box}{\Box}$)** key. Then press **enter**.

20
(2nd) **($\frac{\Box}{\Box}$)** **(enter)**

150÷20%	750
---------	-----

The original cost of the stereo appears on the right of the display: **\$750**.

Practice 4

A. Solve. You MAY use your calculator.

- Find 46% of \$130.
- 14% of what amount is \$13.44?
- What is 12% of \$126?
- What number is $62\frac{1}{2}\%$ of 64?
- 12 is what percent of 400?
- A number is 32% of 6500. What is the number?
- 36 is what percent of 144?
- 90% of what number is 63?
- Find 7% of 360.

- What number is $33\frac{1}{3}\%$ of 237?
- 25 is what percent of 1000?
- What is $12\frac{1}{2}\%$ of 384?
- 390 is what percent of 500?
- Find 2% of \$800.
- 32% of what number is 80?
- What number is $87\frac{1}{2}\%$ of 16?
- \$112 is what percent of \$1600?
- A number is $66\frac{2}{3}\%$ of 414. What is the number?

B. Choose the one best answer to each question. You MAY use your calculator.

Questions 19 and 20 refer to the following chart.

Shipping and Handling Information	
For items costing:	% of order + handling:
\$20 or less	3% + \$1.50
\$20.01 to \$50	4% + \$2.50
\$50.01 to \$100	5% + \$4.00
\$100.01 or more	8%

- Chanel placed an \$84 order. How much shipping and handling will she be charged on her order?
 - \$4.20
 - \$4.40
 - \$8.20
 - \$8.36
- Jason placed an order totaling \$110. Zola placed a \$90 order. How much more did Jason pay in shipping and handling than Zola?
 - \$0.30
 - \$4.30
 - \$4.50
 - \$20.00

- In an election, 3190 out of 3625 registered voters voted against a tax increase. What percent of the registered voters voted against the increase?
 - 43%
 - 83%
 - 88%
 - 98%

- A baseball player has the following statistics. To the nearest percent, what percent of the player's at bats were strikeouts?

At Bats	Hits	Home Runs	Walks	Strikeouts
410	108	2	70	63

- 90%
- 58%
- 26%
- 15%

Answers and explanations begin on page 665.

Lesson 5: Simple Interest

Key Ideas

- Simple interest is found by multiplying principal, rate, and time: $I = prt$.
- The time must be written in terms of years. Write months as a fraction of a year.
- To find the amount paid back, add the principal and the interest.

Interest is a fee paid for the use of someone else's money. If you put money in a savings account, you receive interest from the bank. If you borrow money, you pay interest. In each case, the amount that you invest or borrow is called the **principal**.

Simple interest is a percent of the principal multiplied by the length of the loan. The formula for finding simple interest is given on the formulas sheet of the *Mathematical Reasoning Test* (see page 413). It reads:

$$\text{simple interest} = \text{principal} \times \text{rate} \times \text{time}$$

You may find it easier to remember the formula using **variables** (letters that stand for numbers): $I = prt$, where I = interest, p = principal, r = rate, and t = time. Writing variables next to each other means that they are to be multiplied.

Example 1: Asher borrows \$2500 from his uncle for three years at 6% simple interest. How much interest will he pay on the loan?

1. Write the rate as a decimal.

$$6\% = 0.06$$

2. Substitute the known values in the formula. Multiply.

$$\begin{aligned} I &= prt \\ &= \$2500 \times 0.06 \times 3 \\ &= \$450 \end{aligned}$$

Asher will pay **\$450** in interest.

Some problems ask you to find the **amount paid back**. This adds an additional step to an interest problem. In Example 1, Asher will owe \$450 in interest at the end of three years. However, he will pay back the interest (\$450) plus the principal (\$2500): $\$2500 + \$450 = \$2950$. When he has finished paying the loan, Asher will have paid his uncle \$2950.

In a simple interest problem, the rate is an annual, or yearly, rate. Therefore, the time must also be expressed in years.

Example 2: Eva invests \$3000 for 9 months. She will be paid 8% simple interest on her investment. How much interest will she earn?

1. Write the rate as a decimal.

$$8\% = 0.08$$

2. Express the time as a fraction of a year by writing the length of time in months over 12, the number of months in a year.

$$9 \text{ months} = \frac{9}{12} = \frac{3}{4} \text{ year}$$

3. Multiply.

$$\begin{aligned} I &= prt \\ &= \$3000 \times 0.08 \times \frac{3}{4} \\ &= \$180 \end{aligned}$$

Eva will earn **\$180** in interest.

GED® TEST TIP

An onscreen formula sheet (see page 413) is provided when you take the Mathematical Reasoning Test. The formula for simple interest is printed close to the bottom of that sheet. Every time you see a formula in this book, check to see if it is on the formula sheet. If the formula is not there, you will need to memorize it.

Practice 5

A. Solve these problems using the formula for simple interest. You MAY use your calculator.

1. Leah borrows \$1500 for 2 years at a 12% interest rate. Find the interest on the loan.
2. How much interest would you pay on a loan of \$800 for 6 months at 14% interest?
3. Dominica invested \$2000 for 3 years at an interest rate of 7%. How much interest did she earn on her money?
4. How much interest would you earn on an investment of \$600 for 8 years at 10% interest?
5. Todd invests \$6500 for $3\frac{1}{2}$ years at 5% interest. How much interest will he be paid at the end of the time period?
6. Ricardo borrows \$1850 for 8 months at 12% interest. What is the total amount he will pay back at the end of the loan period?
7. Yanira puts \$5000 in an investment account for 4 years. If she is paid $8\frac{1}{2}\%$ simple interest, how much interest will she earn?

B. Choose the one best answer to each question. You MAY use your calculator.

8. Jean borrowed \$1300 to buy tools for her job as an auto mechanic. The loan is for 1 year 6 months at 9 percent simple interest. Which of the following expressions could be used to find the amount she will pay back at the end of the loan period?

A. $\$1300 \times 1.5 \times 0.09$
 B. $\$1300 \times 1.5 \times 9$
 C. $\$1300 + (\$1300 \times 9 \times 1.5)$
 D. $\$1300 + (\$1300 \times 0.09 \times 1.5)$

9. Noah borrows \$8000 for 5 years to make improvements to his home office. If the simple interest rate is 6 percent, how much will he pay in interest?

A. \$2400
 B. \$1040
 C. \$420
 D. \$400

10. Caleb borrowed \$1500 from his aunt. He plans to pay his aunt back in 9 months. If he pays 4 percent interest on the loan, what is the total amount he will pay back in 9 months?

A. \$1455
 B. \$1545
 C. \$1560
 D. \$2040

Questions 11 and 12 refer to the following information.

Option	Length of Loan	Simple Interest Rate
A	$2\frac{1}{2}$ years	12%
B	3 years	10%
C	4 years	9%

11. Charlotte needs to borrow \$2400. She is considering the three loan options shown above. How much more interest would Charlotte pay if she takes loan option C instead of option A?

A. \$108
 B. \$144
 C. \$720
 D. \$864

12. Charlotte chooses Option B, but she decides to borrow \$2800. What is the total amount she will pay back when the loan is due?

A. \$2830
 B. \$3240
 C. \$3556
 D. \$3640

Answers and explanations begin on page 666.

Lesson 6: Percent of Change

Percent is often used to show change.

Example 1: Michelle recently started her own business. Last month, she earned \$1000. This month, she earned \$2000. How could she describe the increase in her earnings?

Key Ideas

- The amount of change is the difference between the new number and the original number.
- Find the percent of change by dividing the amount of change by the original number.
- Percent of increase may be greater than 100%.

All of the following statements accurately describe the change.

- Michelle's earnings doubled from last month to this month.
- This month, her earnings increased by 100%.
- This month's earnings are 200% of last month's earnings.

Percent of change compares a new number, which shows an **increase** or a **decrease**, to the original number—the number before the change.

Example 2: Before her raise, Lisa earned \$10.50 per hour. Now she earns \$11.34 per hour. What percent raise did her boss give her?

1. Subtract to find the amount of change. $\$11.34 - \$10.50 = \$0.84$
2. Divide the amount of change by \$10.50, Lisa's wage before the change. Convert the decimal to a percent. Lisa's hourly wage increased by 8%.
$$\frac{\$0.84}{\$10.50} = 0.08 = 8\%$$

Think carefully about a situation to decide which number is the original amount.

Example 3: A jacket is on sale for \$90. Three days ago, the jacket was on sale for \$120. By what percent was the price of the jacket reduced?

1. Subtract to find the amount of change. $\$120 - \$90 = \$30$
2. The price of the jacket was \$120 before it was \$90, so \$120 is the original price. Divide the amount of change by \$120. The new price is 25% less than the price three days ago.
$$\frac{\$30}{\$120} = 0.25 = 25\%$$

Percent of increase may be greater than 100%.

Example 4: Calvin started his business with 10 employees. Now he has 60 employees. By what percent has his workforce increased?

1. Subtract to find the amount of change. $60 - 10 = 50$
2. Divide by the original number. Convert the number to a percent. Calvin's workforce has increased by 500%.
$$\frac{50}{10} = 5.0 = 500\%$$

GED® TEST TIP

You can work backward to check your answers. For example, if a price has decreased by 25%, the new price should be 75% of the original price, since $25\% + 75\% = 100\%$.

Practice 6

A. Solve as directed. If necessary, round your answer to the nearest percent. You MAY use your calculator for questions 7 through 10.

- Find the percent of increase from 2000 to 3000.
- Find the percent of decrease from \$2.00 to \$1.25.
- What is the percent of increase from 30 to 90?
- Find the percent of decrease from 20 to 11.
- Find the percent of increase from \$25 to \$30.
- What is the percent of decrease from 500 to 340?
- Find the percent of increase from \$1.89 to \$2.29.
- What is the percent of decrease from 21 to 3?
- Find the percent of increase from 65 to 338.
- What is the percent of decrease from \$1550 to \$1025?

B. Choose the one best answer to each question. You MAY use your calculator.

11. Justin recently moved from a part-time to a full-time job. Because of the change, his weekly pay increased from \$280 to \$448. To the nearest percent, by what percent did his income increase?

A. 38%
B. 60%
C. 168%
D. 267%

12. David bought a computer game on sale for \$36. The game was originally \$48. What was the percent of decrease in the game's price?

A. 12%
B. 25%
C. $33\frac{1}{3}\%$
D. 75%

13. The Utleys' rent increased from \$600 to \$636 per month. By what percent did the rent increase?

A. 4%
B. 5%
C. 6%
D. 7%

Questions 14 and 15 refer to the following information.

Marc sells computer equipment. He buys printers at wholesale and sells them at retail price. Customers who join his discount club pay the member's price.

Printer Pricing Chart

Model Number	Wholesale Price	Retail Price	Member's Price
L310	\$63.00	\$141.75	\$92.15
L1430	\$86.00	\$150.50	\$105.35

14. What is the percent of increase from wholesale to retail price of the L310 model?

A. 56%
B. 78%
C. 125%
D. 225%

15. For the L1430 model, what is the percent of decrease from retail price to member's price?

A. 26%
B. 30%
C. 43%
D. 53%

Answers and explanations begin on page 666.

Ratio, Proportion, and Percent

Practice Questions

Directions: You MAY use your calculator.

- From a total yearly budget of \$360,000, the Kimball Foundation spends \$30,000 on leasing office space. What is the ratio of dollars spent on office space to dollars spent on other costs?
A. 12:1
B. 11:1
C. 1:11
D. 1:12
- A worker can assemble 5 motors in 2 hours. Which of the expressions below could be used to find how long it would take the worker to assemble 50 motors?
A. $2 \times \frac{50}{5}$
B. $\frac{5 \times 50}{2}$
C. $\frac{5}{2 \times 50}$
D. $2 \times 5 \times 50$
- Frank owns a discount music store. The table below shows how much Frank pays for certain merchandise items.

Item	Wholesale Price
CDs	\$7.20
Posters	\$5.60

To find his selling price, Frank increases each price by 35 percent. What is the selling price of a poster?

A. \$9.72
B. \$7.56
C. \$5.95
D. \$1.96
- Neva's car is now worth \$12,000. This is 60 percent of what she paid for it. How much did she pay for the car?
A. \$7,200
B. \$18,000
C. \$19,200
D. \$20,000
- At a shop, the ratio of union to non-union workers is 7 to 3. If there are 18 nonunion workers at the shop, how many union workers are there?
A. 21
B. 25
C. 42
D. 126
- Camilla earned \$954 in commission on \$15,900 in sales. What is her rate of commission?
A. 6%
B. 9%
C. $16\frac{2}{3}\%$
D. 60%
- John spent the following amounts of time building a workbench:
drawing the plans: 2 hours
cutting the wood: $1\frac{1}{2}$ hours
assembling the workbench: 2 hours
sanding and sealing: $3\frac{1}{2}$ hours
What is the ratio of time spent cutting wood to total time spent on the project?
A. 1:9
B. 1:6
C. 1:5
D. 3:7

Questions 8 and 9 refer to the following information.

Ford County Farmland Usage Total Acreage: 40,000	
Usage	Number of Acres
Dairy	22,000
Nursery/greenhouse	3,600
Vegetables/fruits	5,200
Grains	9,200

8. What percent of Ford County farmland is used for the growing of grains, vegetables, or fruits?

A. 23%
B. 36%
C. 57%
D. 64%

9. One dairy farmer in Ford County is considering selling her farm to developers, who will convert it from a dairy farm to a resort. If this happens, the amount of farmland devoted to dairy in Ford County will decrease by 20 percent. How many total acres of farmland will Ford County then have if the dairy farmer decides to sell?

A. 35,600
B. 32,000
C. 17,600
D. 4,400

10. A serving of peanut butter contains 3 grams of saturated fat and 13 grams of unsaturated fat. This amount of fat is 25 percent of the recommended amount of fat in a 2000-calorie diet. What is the ratio of grams of saturated fat to total fat in a serving of peanut butter?

A. $\frac{3}{16}$
B. $\frac{3}{13}$
C. $\frac{13}{16}$
D. $\frac{16}{3}$

11. A drawing of a company logo is 4 inches wide and 5 inches long. If the drawing is enlarged so that it is 12.5 inches long, and the original proportions remain unchanged, how many inches wide will the enlarged drawing be?

A. 7.5
B. 10
C. 15.625
D. 20

12. A local hospital currently has 184 male patients. If the ratio of male to female patients is 4:3, how many female patients are there in the hospital?

Write your answer on the line below.

13. A newspaper advertisement contains the following information.

Busy Body Fitness Center Inventory Reduction Blowout! All sale prices are 20% off original price!	
Equipment	Sale Price
Treadmill	\$1512
Upright bike	\$720
Home gym	\$3148

In dollars, what was the original price of the upright bike?

Write your answer on the line below.

14. The Tigers' ratio of wins to losses is 5 to 4. If the team continues winning at the same rate, how many games will the Tigers win in a 72-game season?

A. 20
B. 40
C. 37
D. 52

15. A television station called 400 adults and asked the following question: "Do you approve of the governor's new education program?" The table below shows the results of the survey:

Response	Percent
Undecided	16%
Yes	32%
No	52%

Of the people called, how many did NOT answer "no"?

A. 64
B. 128
C. 192
D. 208

16. The price of a carton of computer paper decreased from \$25 to \$20. What was the percent decrease in the price?

A. 80%
B. 50%
C. 20%
D. 5%

17. Six months ago, Sandra had 55 regular customers. Now, she has 220 percent as many regular customers as she had six months ago. How many regular customers does Sandra have now?

A. 121
B. 90
C. 66
D. 25

18. If 1 gram of fat equals 9 calories, what percent of the calories in a Munchies roast beef sandwich come from fat?

Munchies Sandwich Facts		
Sandwich	Fat (grams)	Calories
Roast Beef	6	300
Club Classic	5	335

A. 2%
B. 3%
C. 6%
D. 18%

19. For every \$8 in their budget, the Parks spend \$3 on food. If their weekly budget is \$704, how much do they spend on food each week?

A. \$88
B. \$192
C. \$235
D. \$264

20. Suddeth Travel estimates that 80 percent of its employees have more than 12 days of unused sick leave. If 140 employees have more than 12 days of unused sick leave, how many employees work at the agency?

A. 112
B. 164
C. 175
D. 700

21. The Gladstone Theater has 900 seats. At a recent show, the ratio of tickets sold to tickets unsold was 11 to 1. How many tickets were sold?

A. 75
B. 810
C. 818
D. 825

22. Matthew put \$2200 in a savings account for one year and six months. If he earns simple interest at an annual rate of 8 percent, how much will he have in the account at the end of the time period?

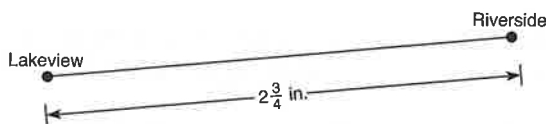
A. \$2212
B. \$2376
C. \$2464
D. \$2640

23. A television set that is regularly priced at \$410 is on sale for 20 percent off. What is the sale price of the television set?

A. \$82
B. \$328
C. \$390
D. \$492

24. On a county map shown below, the map scale reads, "0.5 in. = 60 mi."

What is the actual distance, in miles, between Lakeview and Riverside?



A. 23
B. 165
C. 300
D. 330

Question 25 refers to the following information.

Leo's Bookstore kept track of the number of customers who visited the store over a 3-day period. Employees also recorded the number of customers who made a purchase for each day during the same period.

Day	Number of Customers	Number of Customers Who Made a Purchase
Friday	112	83
Saturday	138	45
Sunday	140	91

25. Which of the following could be used to find what percent of Sunday's customers did NOT make a purchase?

A. $\frac{91}{140}$
B. $\frac{91}{140} \times 100$
C. $\frac{(140 - 91)}{140} \times 100$
D. $\frac{(140 - 91)}{91} \times 100$

26. A school admits 9 out of every 14 who apply. At that rate, how many students will be admitted if 420 apply?

Write your answer on the line below.

27. In a 40-hour workweek, Marcie spends 15 hours answering telephones. What is the ratio of hours spent answering telephones to hours doing other types of work? (Record your answer as a fraction.)

Write your answer on the line below.

Answers and explanations begin on page 666.

CHAPTER 4

Data, Statistics, and Probability

Lesson 1: Tables and Pictographs

Key Ideas

- Tables and graphs are often used to display data.
- Always read all labels to understand what data are presented in the table or graph.
- If symbols are used in a graph, a key will explain what each symbol represents.

Data are facts and information. By analyzing data, we can make predictions, draw conclusions, and solve problems. To be useful, data must be organized in some way. A **table** organizes data in columns and rows. The labels on the table will help you understand what the data mean.

Example 1: The table below shows population figures for selected counties in 2000 and 2010 and the land area in square miles for each county.

County	2000 Pop.	2010 Pop.	Land Area in sq. mi.
Adams	11,128	15,295	4,255
Bell	25,199	22,707	2,523
Cook	6,532	6,518	2,398
Davis	82,204	90,834	1,139
Evans	139,510	130,748	921

Which county showed the greatest percent of increase in population from 2000 to 2010?

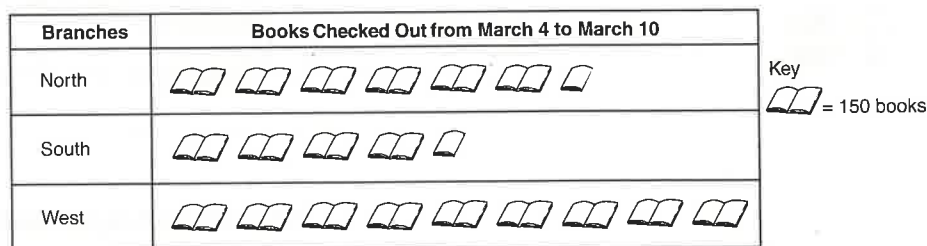
1. **Read the labels.** The first column shows the county names. The second and third columns show population figures. The fourth column shows land area data. You don't need land area to answer this question.
2. **Analyze the data.** Only Adams and Davis counties show increases from 2000 to 2010.
3. **Use the data.** Find the percent of increase for Adams and Davis counties.

Adams: $\frac{15,295 - 11,128}{11,128} \approx 0.374 \approx 37\%$; Davis: $\frac{90,834 - 82,204}{82,204} \approx 0.105 \approx 10\%$

Adams County shows the greatest percent of increase in population from 2000 to 2010.

A **pictograph** is another way to display data. Pictographs use symbols to compare data. A key shows what value each symbol represents.

Example 2: A city has three public library branches. A librarian kept track of the numbers of books checked out from each branch in a week. He used the data to create the pictograph below.



GED® TEST TIP

Tables provide a lot of information. When you are answering a question based on a table, use only the information that is needed to answer the question. In Example 1, the question is about population, so you do not need to use information from the last column.

From March 4 to March 10, how many books were checked out from the South and West branches combined?

- There are $4\frac{1}{2}$ symbols for the South Branch and 9 symbols for the West branch. Add:
 $4\frac{1}{2} + 9 = 13\frac{1}{2}$ symbols.
- Find the value of the symbols. The key states that each symbol equals 150 books. Multiply by 150: $13\frac{1}{2} \times 150 = \mathbf{2025 \text{ books}}$.

Practice 1

A. Use the table on page 280 to answer questions 1 and 2. Use the pictograph on page 280 to answer questions 3 and 4. You MAY use a calculator.

- On average, how many people were there per square mile in Bell County in 2010?
- To the nearest percent, what was the percent of decrease in Evans County's population from 2000 to 2010?
- How many more books were checked out from North Branch than from South Branch during the week of March 4?
- How many books were checked out from all three branches combined?

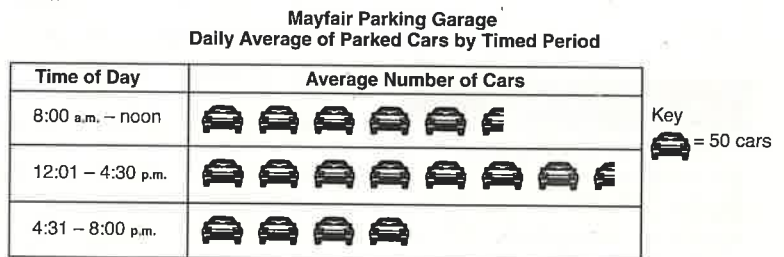
B. Choose the one best answer to each question.

Questions 5 and 6 refer to the following table.

Percent of 3-year-old Children with School-Readiness Skills for the Years 2004 and 2010		
	2004	2010
Recognizes all letters	11%	17%
Counts to 20 or higher	37%	47%
Writes own name	22%	34%
Reads or pretends to read	66%	67%

- If 100 children were surveyed in each year, which category showed the least percent of increase from 2004 to 2010?
 - Recognizes all letters
 - Counts to 20 or higher
 - Writes own name
 - Reads or pretends to read
- A community had 350 three-year-old children in 2010. If the chart is representative of the community, how many were able to write their own name?
 - 34
 - 97
 - 119
 - 134

Questions 7 and 8 refer to the following graph.



- How many cars are parked in the garage from 12:01 to 4:30 p.m.?
 - 275
 - 350
 - 375
 - 650
- How many more cars are parked from 8 a.m. to noon than are parked after 4:30 p.m.?
 - 75
 - 100
 - 175
 - 200

Answers and explanations begin on page 667.

Lesson 2: Bar and Line Graphs

Working with Bar Graphs

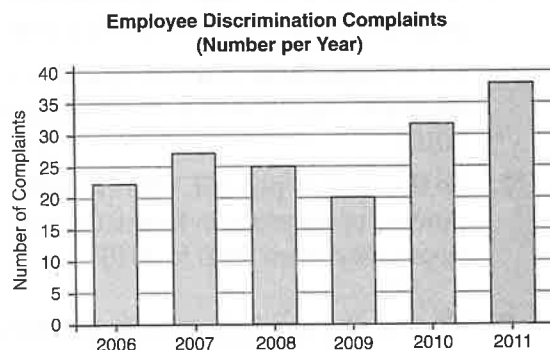
A **bar graph** uses bars to represent values. Bar graphs have two axis lines. One line shows a number scale, and the other shows labels for the bars. By comparing the length of a bar to the scale, you can estimate what value the bar represents.

Key Ideas

- A bar graph uses bars to represent numbers.
- To find the value of a bar, compare its length to the scale shown on one of the axis lines. Estimate the value.
- A double-bar graph compares groups of data. Read the key to find the meaning of the bars.

Example 1: A national corporation made a bar graph (shown below) to show the number of discrimination complaints made by employees during a six-year period. About how many more complaints were made in 2010 than in 2009?

1. **Read the labels.** Each bar represents the number of complaints made within a year. The years are shown beneath the bars.
2. **Analyze the data.** Compare the bars for 2009 and 2010 to the scale. There were 20 complaints in 2009 and about 32 complaints in 2010.



3. **Use the data.** Subtract:
 $32 - 20 = 12$. There were **about 12 more** complaints in 2010 than in 2009.

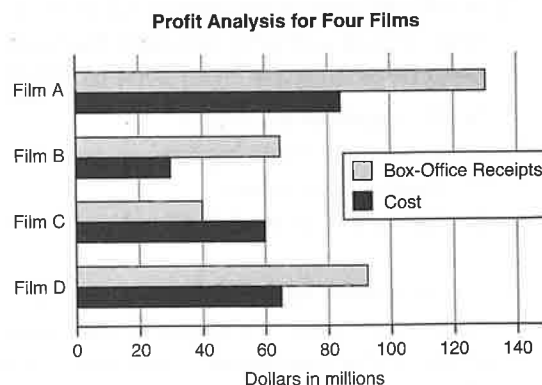
GED® TEST TIP

You may not be able to find the exact value of a bar. If the bar falls about halfway between two numbers on the scale, use a whole number that lies in the middle.

A **double-bar graph** compares more than one type of data.

Example 2: A studio released four films in one year. The graph below compares the cost of making each movie to its box-office receipts, or ticket sales. Film B's cost is what percent of its box-office receipts?

1. **Read the labels.** Read the key to find the meaning of the bars. Notice that the scale represents millions of dollars.
2. **Analyze the data.** Film B's cost is about \$30 million. It brought in about \$65 million in receipts.
3. **Use the data.** Find what percent \$30 is of \$65.
 $\frac{\$30}{\$65} \approx 0.462 \approx 46\%$



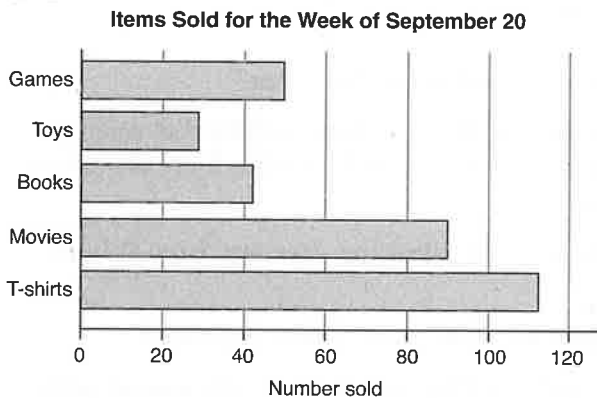
Practice 2.1

A. For questions 1 through 3, use the bar graph entitled “Employee Discrimination Complaints” on page 282. For questions 4 through 6, use the bar graph entitled “Profit Analysis for Four Films” on page 282.

1. To the nearest 10, how many employee discrimination complaints were there in 2006 and 2007?
2. To the nearest 5, how many more complaints were there in 2011 than in 2006?
3. By what percent did the number of complaints decrease from 2008 to 2009?
4. About how much more did it cost to make Film A than Film D?
5. Which film made the greatest amount of profit? (*Hint: Profit = receipts – cost*)
6. Film C’s cost was what percent of its box-office receipts?

B. Choose the one best answer to each question.

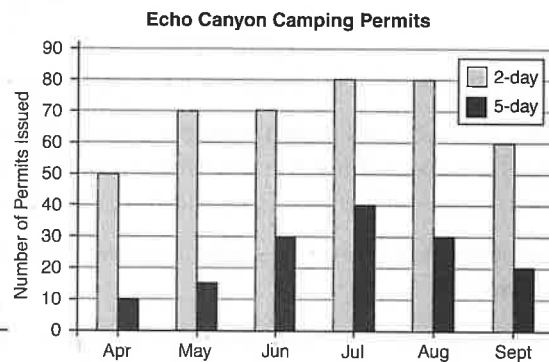
Questions 7 and 8 refer to the following graph.



7. Approximately how many more T-shirts were sold than books and toys combined?
 - A. 40
 - B. 70
 - C. 80
 - D. 90

8. One-half of the games sold during the week of September 20 were on sale for \$16. The rest sold for the full price of \$24. Approximately how much money did the store take in for games sold during the week of September 20?
 - A. \$400
 - B. \$600
 - C. \$800
 - D. \$1000

Questions 9 and 10 refer to the following graph.



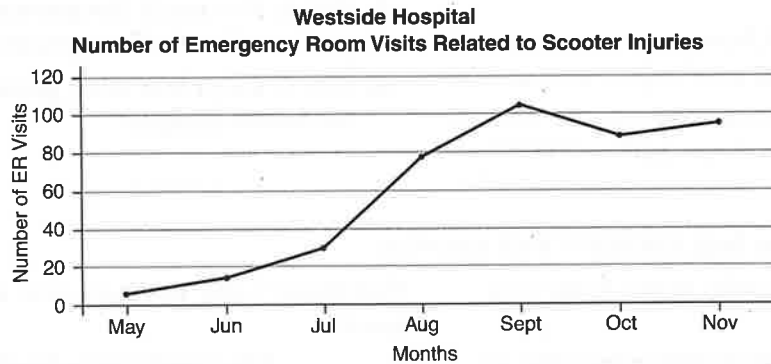
9. In May, what was the ratio of the number of 2-day permits to the number of 5-day permits?
 - A. 2:5
 - B. 3:17
 - C. 14:3
 - D. 14:17
10. In which month was there a total of 80 permits issued?
 - A. June
 - B. July
 - C. August
 - D. September

Answers and explanations begin on page 667.

Working with Line Graphs

A **line graph** is useful for showing changes over time. By analyzing the rise and fall of the line, you can tell whether something is increasing, decreasing, or staying the same. Like a bar graph, a line graph has two axis lines. One is marked with a scale; the other is marked in regular time intervals.

Example 3: The graph below shows the number of patients who visited an emergency room for the treatment of scooter-related injuries.



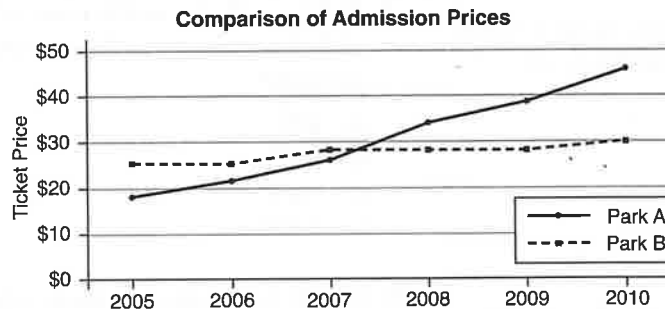
In which month did the greatest increase in scooter-related injuries occur?

The points on the graph are positioned above the months, which are arranged in calendar order. By examining the line that connects the points, you can tell whether there was an increase or decrease from one month to the next.

A steeper line shows a greater increase; therefore, the **greatest increase was from July to August**.

If a line graph has more than one line, a key will tell you what the lines represent.

Example 4: The graph below shows the changes in ticket prices for two amusement parks.



What was the last year in which the admission price to Park B was greater than the admission price to Park A?

The admission prices for Park A are represented by a solid line. Park B's prices are shown with a dotted line. The graph begins in 2005. In 2005, Park B's ticket price is greater than Park A's. Follow the two lines to the right. Between 2007 and 2008, the lines cross, and Park A's prices climb higher than Park B's. **The year 2007** was the last time that Park B charged more than Park A for a ticket.

NOTE: The steepest line shows the greatest increase or decrease, but it may not show the greatest percent of change. When the original value is small, a small change may result in a high percent of change.

Practice 2.2

A. For questions 1 through 3, use the graph from Westside Hospital on page 284. For questions 4 through 6, use the graph "Comparison of Admission Prices" on page 284.

1. In which month did the number of scooter-related injuries decrease?
2. To the nearest 10, how many emergency room visits were due to scooter injuries in August, September, and October combined?
3. Which of the following shows the greater percent of increase: the change in injuries from June to July or the change from August to September?
4. About how much more did it cost to buy a ticket to Park A than a ticket to Park B in 2009?
5. What was the percent of increase in the ticket prices at Park B from 2005 to 2010?
6. To the nearest 10, how much more did it cost to buy a ticket to Park A in 2010 than in 2005?

B. Choose the one best answer to each question.

Questions 7 and 8 refer to the graph below.



7. Over what period of time did the price of goods actually decrease?

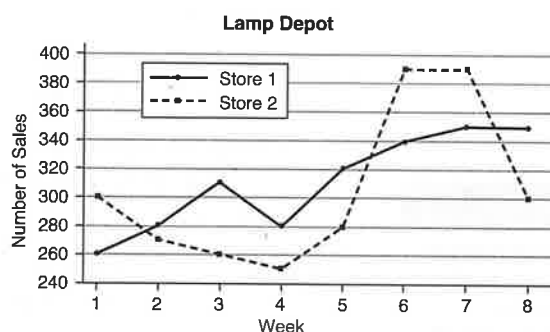
- 1930 to 1940
- 1940 to 1950
- 1960 to 1970
- 1970 to 1980

8. Goods purchased in 1970 were about what fraction of their cost in the year 2000?

- $\frac{4}{5}$
- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{1}{5}$

Questions 9 and 10 refer to the graph below.

Lamp Depot has two stores. The graph shows the sales data from the two stores for an 8-week period.



9. About how many more sales were there at Store 2 than at Store 1 in week 6?

- 110
- 50
- 40
- 25

10. During which week did Store 1 experience the greatest increase in sales from the week before?

- Week 2
- Week 3
- Week 4
- Week 5

Answers and explanations begin on page 667.